

COMPSCI 701:
Creating Maintainable Software
Lecture 3.2: Research on Naming

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Potential Assessment Question

- PAQ

- Agenda
- Preparation
- Class Exercise
- Lawrie et al.
- Hofmeister et al.
- Alsuhaibani et al.
- Key Points

- Martin's heuristic for good names **Use Intention-Revealing Names** says that the name should answer the big questions, including “how the variable, function, or class is used.”

Explain how answering this question improves comprehensibility in terms of the program comprehension model.

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- PAQ
- Admin
 - Assignment 1 questions?
 - Other?
- Wednesday Class Preparation
- Lecture 3.1
- Research on choice of names

Wednesday Class Preparation

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- Create QR Code using Canvas Cover Sheet
- Read the Naming section of Unmaintainable Code
<https://www.mindprod.com/jgloss/unmainnaming.html>

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Class Exercise

- Consider the following possible names for the variable that holds the choice of the row by a player in `NoughtsAndCrosses`:
 - `rowChoice`
 - `rowCh`
 - `rc`

Which is the best? Explain your decision in terms of the program comprehension model

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What the research says (1)

- Dawn Lawrie, Christopher Morrell, Henry Feild, and David Binkley “What’s in a Name? A Study of Identifiers” *14th IEEE International Conference on Program Comprehension (ICPC’06)* 2006.
- Primary hypothesis: full english word identifiers lead to better source comprehension (than abbreviations or single letters)
- Secondary hypotheses: increased work experience and qualifications lead to better ability to comprehend (independent of style of identifier); gender plays a role in confidence but not comprehension
- How to find evidence to support or refute?

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Experimental Artefacts

- two kinds of functions: 6 “text book” algorithms (e.g. binary search); 6 “snippets” taken from production code (e.g. summing debts in an account)
- functions range from 8 LOC to 36 LOC
- each function had 3 variants (**independent variable**: full word(s); abbreviations of words; acronyms (reported as “single letter”). E.g. `isPrimeNumber`, `isPriNum`, `pn`
- C, C++, Java

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Methodology

- Administered by webpage containing Java applet (2006!)
- Demographics: years of computer science education; work experience; job title; age; gender; confidence in languages
- Applet shows two screens for each function: first showing the function, second asking for a description of the purpose of the function and rate their confidence in their description.
- Applet reported time spent on first screen
- Each participant saw one version of each function, selected from the 3 variants to get an even distribution of the independent variable
- Participant sample taken from undergraduates and alumni from Loyola College. Invites by email.

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Results, Sample

- 64 invitees only answered demographics questions
- 128 answered at least one code sample question
- 80 answered all 12 questions
- All code sample results were used whether or not the participant answered all 12 questions

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Results, Data Preparation

- Description correctness independently evaluated by two authors on a scale of 1 (no) to 5 (yes)
 - context implies “yes” means completely correct (not explicitly stated)
 - cohen kappa inter-rater reliability of 0.71 (substantial agreement)
- Data cleaning (age 40+ replaced by 45, unrealistic times, removing incomplete data)

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Results, Statistical Tests

- linear mixed-effects regression
 - simple model with *question*, *variant*, and *question*variant* (interaction)
 - complex model with all demographic information, characteristics of code (e.g. lines of code), measured data (e.g. time)

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Paper Conclusions (selected)

- better comprehension is achieved when full word identifiers are used compared to single letter identifiers (acronyms)
- in many cases abbreviations are as useful as full word identifiers (more true for women than men)
- work experience and education relate to confidence but not to better comprehension

Threats to Validity

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(or, what reasons do we have to doubt the conclusions?)

- unclear whether evaluation of correctness double-blind
- sample taken from Loyola College undergraduates and alumni — how representative of population of all software developers?
- how representative were functions
- perhaps algorithms (as opposed to snippets) can be recognised independent of choice of names
- functions shown in the same order (although names in various combinations)
- **meaningfulness of names**

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Evaluation

- What does our comprehension model predict for this study?
 - How might changes in the KB of the participants explain the study results?
- How are these results useful for improving comprehensibility of code?

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What the research says (2)

- Johannes C. Hofmeister, Janet Siegmund, Daniel V. Holt “Shorter identifier names take longer to comprehend” *Empirical Software Engineering* (2019) 24:417–443
<https://doi.org/10.1007/s10664-018-9621-x> (extension of 2017 paper)
- Replication and extension of Lawrie et al.
- Research Question: *How do identifier name length and semantics affect developers' performance during program comprehension?*
 - RH(Semantic) Words as identifier names lead to faster comprehension of source code than abbreviations and unrelated single letters.
 - RH(Syntax) Identifier naming style has no effect on locating syntactical errors
- Different tasks than Lawrie et al. (semantic and syntax errors)
- Conclusions
 - words as identifiers lead to faster detection of semantic defects
 - when in-depth understanding not required (e.g. syntax errors) identifier names do not make a difference

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What the research says (3)

- Reem S. Alsuhaibani, Christian D. Newman, Michael J. Decker, Michael L. Collard, and Jonathan I. Maletic. 2021. On the Naming of Methods: A Survey of Professional Developers. In Proceedings of the 43rd International Conference on Software Engineering (ICSE '21). IEEE Press, 587–599. <https://doi.org/10.1109/ICSE43902.2021.00061>
- Proposed 10 guidelines for naming identifiers
 - Naming Style—standard such as camel case
 - Grammatical Structure—should be grammatically correct sentences
 - Verb Phrase—should contain verb
 - Dictionary Terms—only natural language words or domain terms
 - Full Words—should be full words
 - Idioms and Slang—should not contain
 - Abbreviations—only known or standard
 - Acronyms—only known or standard
 - Prefix/Suffix—should not contain system-specific prefix/suffix
 - Length—names should not have more than N words (median 5, 90+ $\geq$ 15)
- Survey of 1604 developers asking about level of agreement of guidelines
- Mostly 90%+ agree or strongly agree.
 - Verb phrase—15% disagree/strongly disagree
 - Grammatical structure—21% disagree/strongly disagree
 - Idioms and slang—10% disagree/strongly disagree
 - Prefix/suffix—10% disagree/strongly disagree
- Results only show agreement with proposed guidelines, but say nothing about their effectiveness!

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Key Points

- Research on how to choose names to aid comprehensibility largely supports our intuition, but is not extensive
- Some of the research involves conducting experiments involving human participants performing tasks and measuring their performance
(At UoA, such experiments require permission from the The University of Auckland Human Participants Ethics Committee. Search for “UAHPEC” for details)
- Such research is difficult and expensive